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Design Assumptions vs. Field Reality: A High-Potential Process Safety Incident in Sour Gas Operations

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In recent years, the demand on critical sour gas processing facilities has intensified due to expanded hydrocarbon production, aging infrastructure, and increasingly stringent environmental regulations. Sour gas, which contains significant concentrations of hydrogen sulfide (H_2S) and carbon dioxide (CO_2), poses serious operational, environmental, and safety risks. As oil and gas operators tap into deeper and more complex reservoirs, the volume and toxicity of produced sour gas are increasing, placing added pressure on existing gas processing infrastructure.

Sour gas must undergo extensive treatment before it can be safely transported or used. This includes amine gas treating, dehydration, and sulfur recovery processes each of which must be carefully designed to handle corrosive and toxic conditions. Processing facilities that are not adequately equipped to manage these challenges risk equipment failure, HSE incidents, and environmental violations, all of which carry significant financial and reputational consequences.

One of the most critical yet often underestimated aspects of ensuring long-term, cost-effective performance in sour gas facilities is material selection. The corrosive nature of H_2S and CO_2 , especially in the presence of water, can cause severe corrosion, sulfide stress cracking, and hydrogen embrittlement in metallic components. Poor material choices may result in frequent maintenance, premature equipment replacement, and unplanned shutdowns, all of which escalate operational costs.

To address this, materials must be selected based on their resistance to sour service conditions, mechanical strength, and lifecycle cost. For instance, carbon steel though initially low-cost is highly susceptible to corrosion in sour environments unless coated or protected with inhibitors. In contrast, stainless steels (e.g., 316L), duplex steels, and nickel-based alloys (e.g., Inconel 625) offer superior resistance to sulfide stress cracking and corrosion but come at a higher upfront cost.

Balancing performance and cost requires a thorough understanding of process conditions, including gas composition, pressure, temperature, and moisture content. For example, in low H_2S concentrations or dry gas conditions, corrosion-resistant alloys may not be necessary, and lined carbon steel might suffice. In contrast, high-pressure, wet sour gas applications typically require high-performance alloys to ensure safety and reliability.

Additionally, emerging technologies such as corrosion-resistant composites, internal coatings, and corrosion monitoring systems allow for more flexible and economical approaches to material selection. These can extend equipment life while reducing the need for expensive metallurgical upgrades.

Ultimately, material selection is not a one-size-fits-all decision but a strategic process that influences both capital expenditure (CAPEX) and operational expenditure (OPEX). By aligning material performance with process demands, operators can optimize lifecycle costs, enhance reliability, and ensure regulatory compliance. In the face of increasing throughput and complexity at sour gas facilities, making the right material choices is not just an engineering necessity it is a key enabler of long-term operational and economic success.

Introduction

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Material Selection Considerations in High Sour Gas Facilities: Alloys, Carbon Steel, and Flare System Design

In high sour gas processing facilities, material selection plays a crucial role in ensuring safety, integrity, and cost-effective operation. Sour gas, typically containing high concentrations of hydrogen sulfide (H_2S) and carbon dioxide (CO_2), is highly corrosive and poses serious risks to equipment and personnel. As such, alloy materials such as duplex stainless steels (e.g., 2205), Incoloy 825, or Inconel 625 are commonly specified for sour service environments due to their superior resistance to sulfide stress cracking and general corrosion under wet H_2S conditions.

However, these alloys are significantly more expensive than carbon steel. To optimize costs, design engineers often define material specification breakpoints, particularly in flare and vent systems, where the risk of liquid accumulation is lower. Typically, flare header piping downstream of the sour service section is constructed using carbon steel, provided that operating conditions are dry and the system is designed to prevent liquid hold-up. This is economically advantageous, especially for large-diameter flare headers.

To mitigate the risk of internal corrosion in these carbon steel sections, engineering design incorporates features such as continuous sloping and no-pocket routing, which ensure there is no liquid accumulation. The absence of stagnant liquids reduces the potential for corrosive aqueous phases to form, significantly lowering corrosion risk in carbon steel sections. This approach allows for safe and efficient use of more economical materials without compromising system integrity.

Ultimately, proper material transition points, informed by process conditions and risk assessments, enable the effective balance of safety, durability, and cost in sour gas facilities. Combining appropriate alloy usage in critical areas with smart piping design in lower-risk zones ensures a reliable and cost-effective material strategy.

Challenges in Ensuring Piping Slopes and No-Pocket Design in Sour Gas Facilities: Implications for Corrosion and Long-Term Integrity

In sour gas processing facilities, the importance of maintaining proper slopes and avoiding liquid pockets in piping systems particularly in flare and vent lines is well recognized [Figure-1](#). These design features are critical in preventing the accumulation of corrosive condensates, especially in carbon steel lines where the risk of corrosion under sour conditions is significantly higher. However, in practice, especially during the design and execution of mega projects, achieving these design intentions can be highly challenging.



Figure 1—Pin Hole Leak on the 2" vent line to flare on LP Flash Gas Compressor discharge

During the 3D model review phase, verifying consistent slope across extensive networks of small-bore piping (typically $\leq 2"$) is particularly difficult. These lines are often congested, run in complex routing zones, and may not be clearly visible in overall model views. As a result, minor deviations in slope

or inadvertent low points may go unnoticed until construction or even commissioning stages. The issue is further compounded by the scale of mega projects, where thousands of lines are modeled, making comprehensive manual checks impractical within project timelines.

Additionally, construction limitations in the field such as limited access, tight spaces, or support conflicts can make it physically difficult to install piping with the exact designed slope. In some cases, installers may introduce unintentional pockets or flat spots in small-bore lines, particularly around instrumentation take-offs, drains, or vents. These areas can become dead legs, where liquids condense and accumulate over time.

In high H_2S or CO_2 environments, such stagnant conditions promote localized corrosion, particularly under-deposit corrosion or microbial influenced corrosion (MIC). Over long-term operation, these corroded pockets may eventually lead to gas breakthrough, where corrosive vapors penetrate weakened pipe walls, potentially leading to leaks or catastrophic failure. Such events pose not only safety and environmental risks but also significant costs in terms of shutdowns, repairs, and regulatory compliance.

To mitigate these risks, it is essential to take a proactive approach during both design and construction. On the design side, leveraging automated design rule checks (DRCs) within the 3D model can help identify out-of-slope lines or unintended pockets. Enhanced model reviews that focus specifically on small-bore lines, perhaps using section cuts or isolating vent and flare systems, can also improve detection during design.

From a construction standpoint, clearer documentation and field-friendly slope marking, along with quality inspections focused on slope verification, can help ensure design intent is preserved. Where slope cannot be maintained due to constraints, consideration should be given to alternative mitigation measures such as the use of corrosion-resistant alloys in vulnerable sections, installation of steam or heat tracing to prevent condensation, or providing low-point drains to evacuate accumulated liquids.

In conclusion, while maintaining slope and avoiding pockets may seem like minor design details, they are critical to the long-term integrity of sour service piping systems. Especially in carbon steel flare and vent lines, the consequences of overlooking these details can be severe. A collaborative effort between design, construction, and operations teams is essential to minimize long-term corrosion risk and ensure safe, reliable performance in high sour environments.

Concern Regarding Welding Groove Causing Liquid Accumulation and Corrosion in Vapor Lines

In sour service systems where vapor lines are designed to carry primarily gas-phase hydrocarbons containing hydrogen sulfide (H_2S), the geometry and integrity of the pipeline are critical in minimizing the risk of internal corrosion. A key design principle in such lines is the avoidance of any form of "pockets" or stagnant zones that can allow the accumulation of sour liquid condensate. However, the introduction of an improperly considered welding groove in the pipeline can compromise this design intent.

The welding groove, depending on its geometry, can create a localized low point or a cavity where liquid-phase condensate may accumulate. This is especially problematic in horizontal or sloped vapor lines, where even slight depressions or crevices in the pipe's internal surface can disrupt the self-draining nature of the system. In sour service environments, the gas stream often contains water vapor, hydrocarbons, and acidic gases such as CO_2 and H_2S . Under specific temperature and pressure conditions, especially during startup, cooldown, or process fluctuations, these components may condense into a corrosive liquid phase.

When such condensation occurs, the welding groove becomes a natural trap for the liquid. Since the groove is upstream of the flow direction, the vapor stream may not have enough velocity or turbulence to continuously sweep the groove clean. Over time, this leads to a pooling of sour condensate, forming a localized corrosive environment that is electrochemically different from the rest of the pipeline. This trapped liquid can become stagnant, exacerbating localized corrosion mechanisms such as under-deposit corrosion, crevice corrosion, or pitting.

The design of the vapor line originally accounted for continuous drainage and no liquid hold-up, which is a standard practice in sour service to avoid wet H₂S corrosion. Introducing a welding groove, especially one that disrupts the internal smoothness of the pipe or forms a ledge, contradicts this design philosophy. Moreover, sour service lines often operate at conditions where corrosion mechanisms are accelerated. The presence of even a small amount of standing acidic liquid can initiate localized metal loss, which may remain undetected during routine inspections if the groove is shallow or located in a hard-to-inspect area.

In the long term, this corrosion can weaken the structural integrity of the line, leading to increased maintenance costs, unplanned shutdowns, or in extreme cases, failure of the pipeline. This is particularly dangerous in sour environments due to the toxic and flammable nature of the contents.

To mitigate this risk, it is crucial to adhere strictly to the no-pocket design philosophy. Any proposed welding groove should be carefully evaluated for its internal geometry impact. Alternative welding or joining techniques should be considered if they maintain a smooth, flush internal surface. Additionally, any modification to the line should undergo a rigorous corrosion review and internal flow simulation to ensure that condensate accumulation does not occur.

In conclusion, although welding grooves may be necessary for structural or fabrication reasons, their implications on sour service lines must be thoroughly assessed. The potential for such features to trap corrosive liquids and compromise the corrosion resistance of the line should not be underestimated.

Justification for Removal of Welding Groove and Implementation of Proper Sloping in Sour Service Vapor Lines

During the construction phase of sour service vapor lines, it is critical to align all fabrication activities with the original design intent particularly when the design assumes a "no pockets" configuration to justify material selection. The introduction of a welding groove at this stage not only deviates from this intent but also poses a significant long-term corrosion risk, especially in lines handling sour vapors containing hydrogen sulfide (H₂S) and other corrosive components.

The original design of the vapor line considered sloped routing with no pockets, specifically to prevent liquid accumulation and thereby minimize internal corrosion risk. This "no-pockets" approach was a fundamental criterion in material specification and corrosion allowance decisions. In many sour service applications, materials such as carbon steel are deemed acceptable only under the assumption that liquid-phase sour condensates will not be allowed to accumulate, thereby limiting corrosion exposure. This credit is commonly referred to as the "no-liquid-hold-up" or "no-pockets" assumption in material selection philosophy.

Introducing a welding groove, particularly one that results in a local depression or crevice inside the pipe wall, violates this assumption. The groove can unintentionally act as a liquid trap, creating a microenvironment conducive to corrosive attack. Over time, sour condensates formed through phase change due to cooling, pressure drops, or flow interruptions can collect in such grooves. Even minor pooling in sour service systems can initiate localized corrosion mechanisms, including sulfide stress cracking, pitting, and under-deposit corrosion. This localized damage may go undetected until advanced, especially if it occurs in less accessible areas.

Moreover, retaining the groove contradicts the project's corrosion control strategy, which assumed no liquid hold-up in the vapor line and justified the use of non-corrosion-resistant alloy materials. By allowing features that create stagnant zones, the corrosion risk becomes inconsistent with the material selection basis, and the system's reliability is compromised.

To mitigate this risk, it is strongly recommended that the welding groove be removed during the construction phase. All welds should be executed in a manner that ensures a flush internal bore, matching the pipe inner diameter and preserving smooth flow conditions. Alternatives such as full-penetration welds with internal alignment rings or back grinding may be considered to eliminate internal discontinuities.

In addition, it is essential to reinforce proper sloping throughout the vapor line. Each segment should be verified to ensure that it drains naturally without any low points or stagnant zones. Field inspections during construction should verify slope continuity and the absence of fabrication-related anomalies that could form unintended pockets.

In conclusion, preserving the design intent of "no pockets" is not merely a best practice—it is a technical requirement linked directly to material performance and service life in sour environments. Removing the welding groove and ensuring proper slopes will uphold the integrity of the system, avoid premature corrosion issues, and remain compliant with the assumptions made during design and material selection.

Recommendations

To mitigate the risks associated with condensation-induced corrosion in sour gas systems, a comprehensive set of recommendations has been developed. These recommendations aim to improve design integrity, material selection, and operational safety. The focus is on identifying vulnerabilities caused by ambient cooling, dead legs, and material suitability, especially in vent and discharge lines. The following actions are proposed:

- Engage with Process Engineering, Materials and Corrosion, and Mechanical CFDH's to enhance the design procedures with an assessment of risks associated with ambient cooling in processes where condensation may occur.
- Complete a PDO-wide assessment of compressor discharge systems to ensure all dead legs are safe from condensation-induced cooling and corrosion. This includes identifying and reviewing stagnant long lines, with an expanded definition of dead legs.
- Assess and revise the existing design to ensure it is suitable for the actual service conditions, and rectify any identified issues (e.g., P-31008 line) through proper FCP and execution.
- Conduct a thorough evaluation of all area compressors to determine whether similar risks exist across other equipment and systems.
- Perform a detailed metallurgical analysis to confirm the exact corrosion mechanism behind the observed failure.
- Undertake a risk assessment to define the risks and necessary controls for operating carbon steel (CS) lines until a permanent solution is implemented.
- Review and revise the material selection criteria for vent and bypass lines to ensure compatibility with potential wet sour gas service, especially when their design conditions differ from the main process lines and may promote condensation.

Conclusion

As sour gas facilities continue to evolve in scale, complexity, and environmental sensitivity, the challenges associated with safely and efficiently handling corrosive fluids have intensified. These facilities are increasingly required to process high volumes of hydrogen sulfide (H_2S) and carbon dioxide (CO_2), often under high-pressure, high-temperature conditions. Under such circumstances, material selection and piping design become critical factors in safeguarding asset integrity, protecting personnel, and ensuring regulatory compliance.

One of the central strategies to mitigate corrosion risk in sour gas systems is the thoughtful and intentional application of material selection principles. While high-performance alloys like duplex stainless steels, Incoloy 825, and Inconel 625 offer superior corrosion resistance, their high cost necessitates a balanced approach. Carbon steel remains widely used due to its cost-effectiveness but is only viable under specific assumptions chief among them being the "no pockets" or "no liquid hold-up" condition. This design assumption underpins material specifications, corrosion allowances, and long-term performance

projections. Any deviation from this principle whether through construction oversights, design lapses, or field modifications can compromise the entire integrity philosophy of the system.

The installation of welding grooves in vapor lines is one such deviation that poses a significant and often underappreciated threat. A seemingly minor structural feature, the groove can act as a micro-trap for sour condensates. These grooves disrupt the smooth internal profile of the pipe and create localized stagnation zones where water, hydrocarbons, and acid gases can condense and collect. Over time, this leads to the formation of highly corrosive environments that accelerate sulfide stress cracking, under-deposit corrosion, and localized wall thinning especially in carbon steel systems. The issue becomes even more concerning when these grooves remain upstream of the flow direction, where they are less likely to be self-scavenged by gas velocity.

The risk is not merely theoretical. In practice, such features have led to premature pipeline failures, production losses, and safety incidents. Their presence violates the very basis upon which carbon steel was selected, rendering corrosion mitigation strategies ineffective and non-compliant with engineering assumptions documented during the FEED and detailed design phases.

Furthermore, achieving proper slope and no-pocket design is not just a drawing-board requirement it must be vigorously enforced during construction. Mega projects with extensive small bore piping networks face real-world challenges in maintaining consistent slope due to congested layouts, limited visibility during 3D model reviews, and physical constraints in the field. Instrument take-offs, drains, vents, and supports often introduce unintended low points. These inconsistencies, though minor in appearance, can become major points of corrosion during operation.

Addressing these challenges requires a multi-pronged approach. During the design phase, leveraging advanced 3D design rule checks and isolating critical systems such as flare and vent lines for focused slope reviews can minimize unintentional pockets. During construction, welders and inspectors must be trained and empowered to identify and eliminate internal discontinuities, such as grooves, that deviate from the design intent. Quality checks must include not just dimensional compliance but also hydraulic and corrosion implications of minor imperfections. Where slope cannot be guaranteed due to space or routing limitations, alternative mitigations such as corrosion-resistant alloy inserts, heat tracing, or drain points should be proactively implemented.

Ultimately, preserving the "no-pockets" design philosophy is not just a matter of engineering rigor it is a contractual, safety, and operational obligation. It maintains the integrity of material selections, upholds the economic assumptions behind CAPEX and OPEX models, and directly contributes to the safe handling of some of the most hazardous fluids in the oil and gas industry. Every stakeholder from designers to field engineers and inspectors plays a role in ensuring that this principle is honored throughout the project lifecycle.

The removal of unnecessary welding grooves and strict adherence to sloping standards are low-cost, high impact actions that can dramatically extend asset life, reduce corrosion-related failures, and optimize the total cost of ownership. As sour gas processing facilities become more integral to global energy supply chains, engineering decisions down to the smallest weld detail must reflect a holistic understanding of risk, performance, and lifecycle value.

In conclusion, the immediate cause of the vent line failure was internal pitting corrosion resulting in a pinhole leak. The line was fabricated from carbon steel, assuming dry gas conditions based on the main 10" flash gas line. However, the 2" vent line experienced stagnant flow and significant cooling, which led to condensation of free water a condition not anticipated during the original design phase. This created an internal wet sour environment, accelerating corrosion and ultimately causing the leak. The investigation identified a key systemic issue: the direct extrapolation of process conditions from the primary to auxiliary lines without considering differing thermodynamic behaviors, especially in low-flow, uninsulated segments. Additionally, the lack of verification on material compatibility under worst-case conditions led to an inappropriate material selection for a potentially corrosive service. The incident underscores the importance

of re-evaluating secondary system design criteria, improving thermal profiling during FEED, and ensuring that conservative assumptions guide material choices in sour applications. Implemented changes have since improved inspection protocols and revised the design basis for vent and drain lines